Pierrick Le Vourc'h

Temporary Research and Teaching Attaché

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Elvipy

Education

2023–2026 **PhD in mathematics**, *IMAG, Université de Montpellier*, Montpellier, France
Thesis untitled *Derivation and analysis of models for compressible multiphase flows*, supervised by Nicolas Seguin and Khaled Saleh.

2019–2023 **École Normale Supérieure de Lyon**

- Master of Science in Advanced mathematics, PDE major (2022–2023)
- Agrégation de mathématiques (2021–2022)

  French national competitive examination for the recruitment of teachers in mathematics.
- Master 1: Advanced mathematics (2020–2021)
- Bachelor of Science in Fundamental Mathematics (2019–2020)

2016–2019 **CPGE MPSI-MP***, *Lycée La Pérouse-Kerichen*, Brest, France
French preparatory program to prepare the competitive exams for engineering and top scientific studies. Option: computer sciences.

Teaching and scientific mediation

Teaching experience

2026–2027 **Temporary Research and Teaching Attaché**, *Université de Montpellier*, Montpellier, France
Seminar teacher for college students (192h). Supervision of mathematics exercise sessions in biology, computer science, engineering and mathematics bachelors. Preparation for french competitive examinations for recruitment of teachers.

2025–2026 **Seminar teacher**, *Université de Montpellier*, Montpellier, France
Supervision of mathematics exercise sessions in computer science, engineering and mathematics bachelors (64h).

2024–2025 **Seminar teacher**, *Université de Montpellier*, Montpellier, France
Supervision of mathematics exercise sessions in biology and mathematics bachelors (64h).

2023–2024 **Seminar teacher**, *Université Claude Bernard Lyon 1*, Lyon, France
Supervision of mathematics exercise sessions for BTEC students in mechanical engineering and mathematics bachelor students (64h).

2022–2023 **Supervisor of oral evaluations in preparatory classes**, *Lycée du Parc*, Lyon, France

2020–2021 **Mentorship**, *ENS de Lyon*, Lyon, France
Mentorship of a college freshman and support for high school students on Saturday afternoons.

Scientific mediation

2026 **Science for all**, Montpellier, France
Creation and animation of a cryptography workshop for high school students.

2025 **Science festival**, Montpellier, France
Presentation of the exhibition *Why mathematics?* and exchanges with young people in the Science van Carasciences.

2025 **Jury for the Young Mathematicians French Tournament (TFJMM)**, Montpellier, France
Evaluation of written solutions to various problems. Evaluation of oral presentations and debate sessions throughout the weekend.

2023–2025 **Author of competitive examinations corrections**, *H&K Editions*, France
Writing and proofreading of corrections of mathematics papers of engineering schools' competitive exams.
Find the books I contributed to at h-k.fr/pierrick.le-vourc-h.

2023, 2024 **MathαLyon**, Lyon, France
Animation of different workshops for middle and high school students.

Responsibilities

2024–2026 **Member of the parity commission**, *IMAG*, Montpellier, France

2024–2025 **Co-organizer of the PhD students' seminar**, *IMAG*, Montpellier, France

2024 **Co-organizer of the CJC-MA24 congress**, *UMPA & ICJ*, Lyon, France
Congress of Young Researchers in Applied Mathematics
Website: `https://cjc-ma2024.sciencesconf.org/`

Research activities

Preprints

2026 **Averaged models for compressible two-phase stratified flows on thin domains**
P. Le Vourc'h, K. Saleh, N. Seguin
Lien HAL: `https://hal.science/hal-05096837v2`

Talks and posters

06/2026 **ProHyp26**, Strasbourg, France
Presentation of a talk untitled *Mathematical analysis and numerical simulation of a Kapila model*.

05/2026 **HYP26**, Stuttgart, Allemagne
Presentation of a talk untitled *Formal derivation of a stratified compressible two-phase flow model*.

10/2025 **PhD students' seminar**, *IMAG*, Montpellier, France
Presentation of a talk untitled *An introduction to the modelization of multiphase flows*.

05/2025 **7th Workshop on Compressible Multiphase Flows**, Strasbourg, France
Presentation of a talk untitled *Derivation of averaged compressible two-phase flow models*.

12/2024 **Regional PhD students' seminar**, Toulouse, France
Presentation of a poster.

11/2024 **Scientific days of the CNRS Network *Terre & Énergies***, Nouan-Le-Fuzelier, France
Presentation of a poster.

06/2024 **6th Workshop on Compressible Multiphase Flows**, Strasbourg, France
Presentation of a poster.

05/2024 **46th National Congress on Numerical Analysis**, Île de Ré, France
Presentation of a poster.

02/2024 **PhD students' seminar**, *ICJ*, Lyon, France
Presentation of a talk untitled *Modèles moyennés pour les écoulements multiphasiques compressibles*.

01/2024 **Integration Days of the Institut Camille Jordan**, Lyon, France
Presentation of a poster.

Skills

Computer skills

- Programming: Python, shell, git, Fortran
- Softwares: LaTeX, Suite Office
- Operating systems: GNU/Linux (Debian, Ubuntu), Windows

Languages

- French: mother tongue
- Breton: mother tongue
- English: C2 level (Cambridge English Advanced, 205 points)
- Spanish: intermediate level

Interests

Cooking ▪ Sports (working out, running, rock climbing) ▪ Board games ▪ Knitting